CE-08: Dense Negative Control

Two-Layer CTMC Experiments Pipeline

August 4, 2026

Abstract

CE-08 tests whether the large sparse-community discrepancy persists in a short, well-mixed clique chain. Across 200 paired replications, completion was identical and the RMST shift was small with a confidence interval spanning zero. The negative control therefore behaves as expected: the approximation does not exhibit a detectable systematic timing bias when internal mixing is fast and the chain is short. This supports, but does not prove, the boundary-reachability mechanism proposed for sparse communities.

1 Design and results

The network contains $L = 3$ cliques of size $n = 100$, $k = 2$, $\beta_{in} = \beta_{out} = 0.003$, and $T_{end} = 1500$. Both models completed all 200 runs.

p_M	p_{MM}	ΔRMST	95% CI	rel-W_1	KS p
1.00	1.00	8.20	$[-38.99, 54.45]$	0.0647	0.545

Table 1: Terminal-arrival comparison; differences are MicroMacro minus Full Micro.

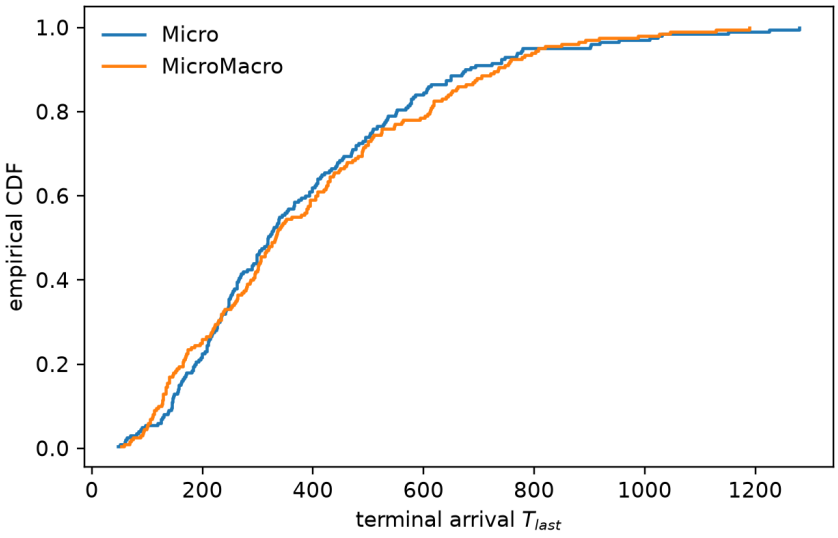


Figure 1: Terminal empirical CDFs. The independent statistical unit is a paired replication.

2 Interpretation and implications

The RMST estimate is only 0.55% of the horizon and its interval includes zero. This is evidence against a universal scheduler offset large enough to explain the sparse-chain results. A correction

intended for sparse communities should therefore self-disable on clique-like or rapidly mixing communities; otherwise it risks introducing error where the baseline already agrees.

Failure to reject a difference is not proof of equivalence because no predeclared equivalence margin was used. The result is nevertheless a useful falsifiability check: the same comparison pipeline does not mechanically report a large premature bias in every topology.

Reproduction

```
.venv/bin/python extrascripts/run_ce08_negative_control.py -grid full -workers 12
-resume
```